

Refraction at Curved Surfaces, Thin Lenses, and the Simple Magnifier

PHYS 2203 — Introductory Physics III

September 9, 2026

Overview

This lecture turns from mirrors to image formation by *refraction*. We first analyze refraction at a single spherical surface, chain two such surfaces through a glass sphere, and then collapse that chain into a single element — the *thin lens*. We close by using a converging lens as a *simple magnifier*, forming an enlarged virtual image of a nearby object. The next lecture brings in the eye, introduces *angular magnification*, and combines lenses in series to build the compound microscope and the refracting telescope.

1 Refraction at a Single Spherical Surface

The previous lecture's mirror analysis used reflection. The identical program — paraxial rays, an image equation, sign conventions — applies when light *refracts* at a curved boundary. Consider an interface between two transparent media of refractive indices n_1 and n_2 , where the boundary is a spherical surface of radius of curvature r . An object placed in medium 1 at distance S from the vertex is imaged by refraction at the surface, forming an image in medium 2 at distance S' .

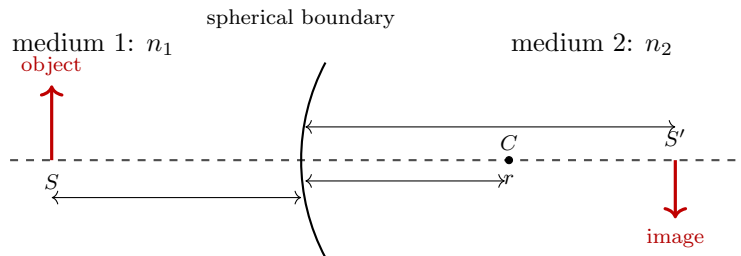


Figure 1: Refraction at a single spherical surface separating two media of refractive indices n_1 and n_2 . An object in medium 1 is imaged in medium 2 at signed distance S' from the vertex.

Spherical-refraction equation

Applying Snell's law to paraxial rays ($\sin \theta \approx \theta$), the object distance, image distance, refractive indices, and radius of curvature are related by

$$\boxed{\frac{n_1}{S} + \frac{n_2}{S'} = \frac{n_2 - n_1}{r}} \quad (1)$$

and the transverse magnification is

$$m = -\frac{n_1}{n_2} \frac{S'}{S}. \quad (2)$$

Note the family resemblance to the mirror equation: reciprocal distances on the left, a curvature-driven constant on the right. (A derivation of the refraction equation from Snell's law is given in the optional section at the end of these notes.)

Sign conventions

- $S > 0$ for an object on the side of the surface from which light is incident.
- $S' > 0$ when the image is on the transmission side of the surface (real image); $S' < 0$ when the image is on the incident side (virtual image).
- $r > 0$ for a *convex* surface, i.e. when the center of curvature C lies on the transmission side (medium 2).
- $r < 0$ for a *concave* surface, i.e. when C lies on the incident side (medium 1).

With these conventions the equation handles every combination of curvature and indices uniformly.

2 Worked Example: Hummingbird Viewed Through a Glass Sphere

A hummingbird hovers at a distance of 50 cm from a glass sphere of radius 50 cm. Determine the location of the image as seen by an observer on the opposite side of the sphere.

The geometry comprises two refracting surfaces: an entrance surface (air→glass) and an exit surface (glass→air). The two are treated in sequence, with *the image of the first surface acting as the object for the second* — a principle that recurs throughout optics.

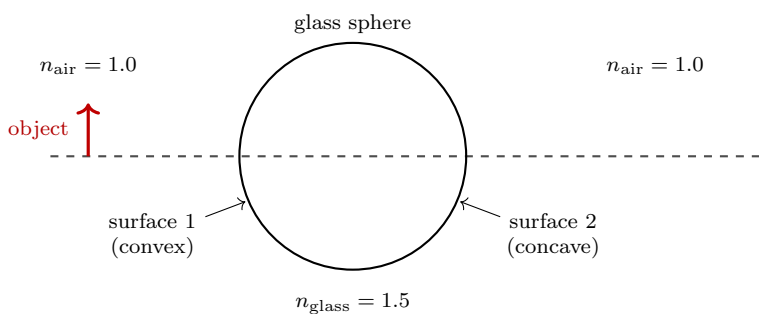


Figure 2: A glass sphere (radius 50 cm, $n = 1.5$) imaged by a hummingbird at 50 cm to its left. Surface 1 is the air→glass interface; surface 2 is the glass→air interface.

Surface 1 (air → glass)

Given. $S_1 = 50$ cm, $r_1 = +50$ cm (convex), $n_1 = 1.0$ (air), $n_2 = 1.5$ (glass).

Image distance. Applying the spherical-refraction equation,

$$\begin{aligned}\frac{1.0}{50 \text{ cm}} + \frac{1.5}{S'_1} &= \frac{1.5 - 1.0}{50 \text{ cm}} = 0.01 \text{ cm}^{-1}, \\ \frac{1.5}{S'_1} &= 0.01 \text{ cm}^{-1} - 0.02 \text{ cm}^{-1} = -0.01 \text{ cm}^{-1}, \\ S'_1 &= \frac{1.5}{-0.01 \text{ cm}^{-1}} = -150 \text{ cm} \quad (\text{virtual}).\end{aligned}$$

Magnification.

$$m_1 = -\frac{n_1}{n_2} \frac{S'_1}{S_1} = -\frac{1.0}{1.5} \frac{-150}{50} = +2 \quad (\text{upright, larger}).$$

The intermediate image is virtual, 150 cm to the left of surface 1, upright, and twice the size of the hummingbird.

Surface 2 (glass $\rightarrow$ air)

The intermediate image acts as the object for the second surface. Its location is 150 cm to the left of surface 1, which is $150 + 100 = 250$ cm to the left of surface 2 (since the sphere has diameter 100 cm).

Given. $S_2 = +250$ cm, $r_2 = -50$ cm (concave when viewed from inside the glass), $n_1 = 1.5$, $n_2 = 1.0$.

Image distance.

$$\begin{aligned}\frac{1.5}{250 \text{ cm}} + \frac{1.0}{S'_2} &= \frac{1.0 - 1.5}{-50 \text{ cm}} = \frac{-0.5}{-50} \text{ cm}^{-1} = 0.01 \text{ cm}^{-1}, \\ \frac{3}{500 \text{ cm}} + \frac{1}{S'_2} &= \frac{5}{500 \text{ cm}}, \\ \frac{1}{S'_2} &= \frac{2}{500 \text{ cm}}, \quad \text{so} \quad S'_2 = \frac{500 \text{ cm}}{2} = 250 \text{ cm} \quad (\text{real}).\end{aligned}$$

Magnification.

$$m_2 = -\frac{n_1}{n_2} \frac{S'_2}{S_2} = -\frac{1.5}{1.0} \frac{250}{250} = -1.5.$$

Final image

The total magnification is the product of the two stages:

$$m = m_1 m_2 = (2)(-1.5) = -3 \quad (\text{inverted, larger}). \quad (3)$$

The observer sees a real image 250 cm to the right of the sphere, inverted and three times the size of the hummingbird.

We now collapse this two-surface analysis into a single element: the *thin lens*.

3 Thin Lenses

The glass-sphere example above was really a lens in disguise: two refracting surfaces traversed in sequence. A **thin lens** is a piece of transparent material (typically glass) bounded by two closely-spaced refracting surfaces. When the lens thickness is much smaller than every other relevant length scale, the two surfaces can be treated as if they coincide at the lens center, and the two-surface chain becomes a single equation.

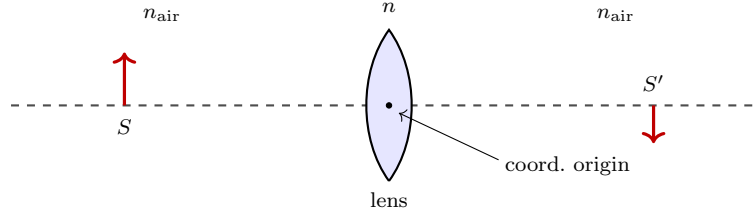


Figure 3: Schematic of a thin lens. Both refracting surfaces are taken to coincide at the lens center, which is the origin for the object distance S and image distance S' .

The thin-lens equation

The object distance, image distance, and focal length f of a thin lens are related by

$$\boxed{\frac{1}{S} + \frac{1}{S'} = \frac{1}{f}} \quad \boxed{m = -\frac{S'}{S}}. \quad (4)$$

These are *identical in form* to the mirror equation and magnification derived in the last lecture. The focal length is defined by $S' = f$ when $S \rightarrow \infty$ (parallel incoming rays). Unlike a mirror, a lens is *traversed* by the light, so it has *two* focal points: parallel rays arriving from one side converge at the **second focal point** F' , a distance f beyond the lens, and parallel rays arriving from the other side converge at the **first focal point** F , a distance f in front of it. The two distances are equal even for a lens with dissimilar faces, because rays are reversible: e.g. the thin-lens equation is unchanged under $S \leftrightarrow S'$.

Lensmaker's equation

The focal length of a thin lens of material refractive index n in air is given by the **lensmaker's equation**:

$$\boxed{\frac{1}{f} = \left(\frac{n - n_{\text{air}}}{n_{\text{air}}} \right) \left(\frac{1}{r_1} - \frac{1}{r_2} \right)}, \quad (5)$$

where r_1 is the radius of curvature of the first surface (the one the light strikes first) and r_2 is that of the second surface. The same sign convention used for a single refracting surface applies: $r > 0$ when the center of curvature lies on the transmission side of the corresponding surface, $r < 0$ otherwise.

The focal length depends on the curvatures r_1 and r_2 , the lens material index n , and the surrounding medium index n_{air} . For lenses immersed in air, $n_{\text{air}} \approx 1$ and $n > 1$, so the sign of f is determined by the geometric factor $1/r_1 - 1/r_2$.

Power of a lens

The reciprocal of the focal length measures how strongly a lens bends parallel light and is called the **power** of the lens. With f expressed in meters, the power is in **diopters** (D):

$$\boxed{P = \frac{1}{f}} \quad 1 \text{ D} = 1 \text{ m}^{-1}. \quad (6)$$

A converging lens with $f = 25 \text{ cm} = 0.25 \text{ m}$ has $P = +4.0 \text{ D}$; a diverging lens has $f < 0$ and hence negative power. Eyeglass prescriptions are quoted in diopters.

Converging versus diverging lenses

- $f > 0$: **converging** lens. Parallel rays incident from one side converge to the focal point on the opposite side.
- $f < 0$: **diverging** lens. Parallel rays incident from one side appear, after the lens, to diverge from a focal point on the same side as the source.

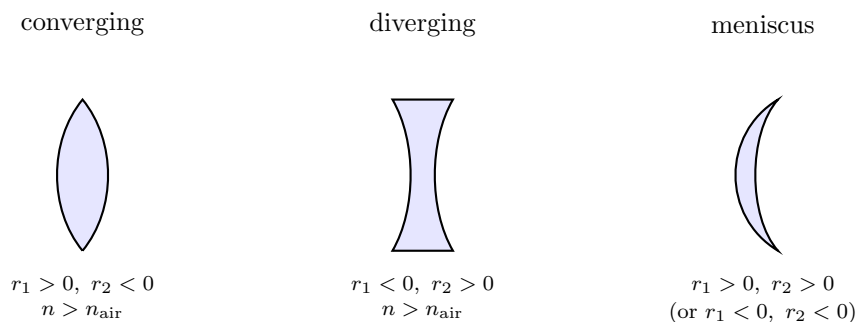


Figure 4: Standard lens shapes. A biconvex lens in air is converging; a biconcave lens is diverging. A meniscus lens (both surfaces curved in the same direction) is converging when its more strongly curved surface is the convex one, and diverging otherwise.

Ray diagrams for thin lenses

As with mirrors, images can be located graphically. In the thin-lens approximation all of the bending occurs at the central plane of the lens, and three principal rays from an object point are easy to trace:

- A ray parallel to the axis emerges directed through the second focal point F' (the definition of F').
- A ray through the center of the lens passes undeviated (the faces are parallel there, and for a thin lens the sideways offset is negligible).
- A ray through the first focal point F emerges parallel to the axis (ray (a) reversed).

Any two of the three rays locate the image. The third serves as a check. For a diverging lens the same three rays are drawn using the focal points on the opposite sides of the lens.

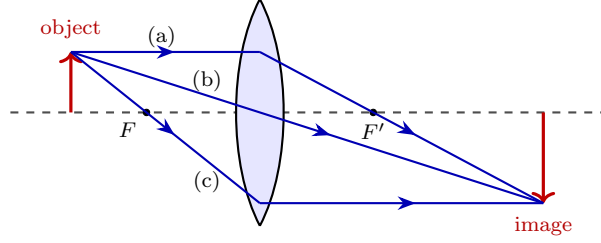


Figure 5: The three principal rays for a thin converging lens, drawn for an object outside the focal length: (a) the parallel ray emerges through F' ; (b) the central ray passes through the lens center undeviated; (c) the focal ray through F emerges parallel to the axis. All three converge on the same image point (real and inverted here).

4 The Simple Magnifier

A converging lens used as a **simple magnifier** produces an enlarged virtual image of an object placed close to it. The condition is that the object distance satisfy $0 < S < f$, where $f > 0$ is the focal length of the lens.

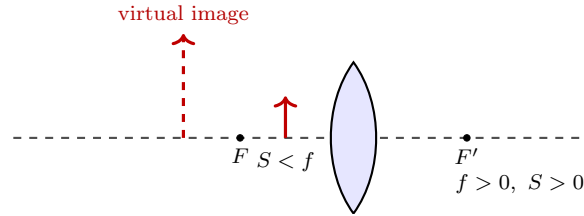


Figure 6: A converging lens used as a magnifier: an object inside the focal length ($S < f$) produces an enlarged, upright, virtual image on the same side as the object.

Quantitative behavior

For a converging lens with $f > 0$, the thin-lens equation gives

$$\frac{1}{S'} = \frac{1}{f} - \frac{1}{S} \implies S' = \frac{fS}{S - f}, \quad m = -\frac{S'}{S} = \frac{f}{f - S}. \quad (7)$$

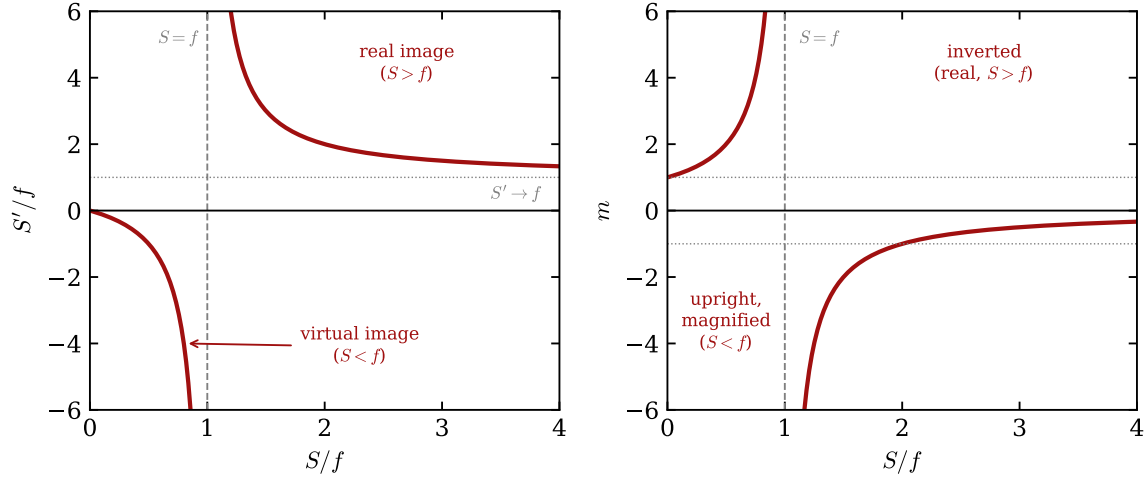


Figure 7: Image distance S' and magnification m as functions of object distance S for a converging lens of focal length $f > 0$, in units of f . Both diverge as $S \rightarrow f$. For $0 < S < f$ the image is virtual, upright, and magnified (the magnifier regime); for $S > f$ it is real and inverted. As $S \rightarrow \infty$, $S' \rightarrow f$.

Summary of the magnifier regime

For $0 < S < f$:

- $S' < 0$: the image is on the same side of the lens as the object (*virtual*).
- $m > 1$: the image is upright and magnified.

The image is observed by looking through the lens at the object. To use a converging lens as a magnifying glass, hold it at a distance less than or equal to its focal length from the object.

Optional: Deriving the Spherical-Refractive Equation

The spherical-refraction equation follows from Snell's law applied to a single paraxial ray, plus triangle geometry. Consider a point object P on the axis at distance S from the vertex, and follow one ray that strikes the surface at a point A , a small height h above the axis. The normal at A is the radius line through C . After refraction the ray crosses the axis at P' , a distance S' from the vertex.

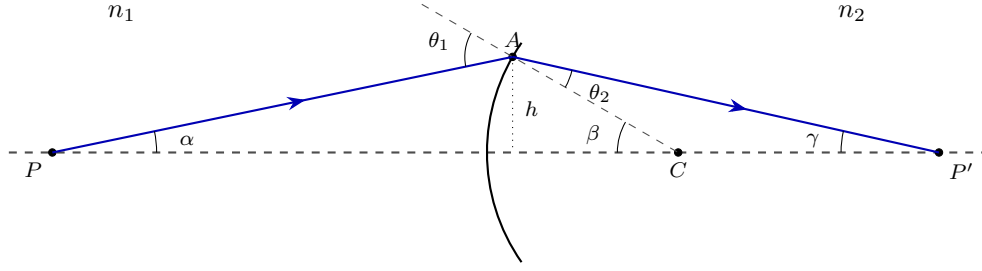


Figure 8: Geometry for the derivation. A single ray from P refracts at A (height h) and crosses the axis at P' . The normal at A is the radius line through C . The angles α , β , γ are measured from the axis at P , C , and P' ; the angles θ_1 , θ_2 are measured from the normal.

Step 1 — exterior angles. In triangle PAC , the incidence angle θ_1 is the exterior angle at A , so $\theta_1 = \alpha + \beta$. In triangle ACP' , β is the exterior angle at C , so $\beta = \theta_2 + \gamma$, i.e. $\theta_2 = \beta - \gamma$.

Step 2 — Snell's law, paraxial. With $\sin \theta \approx \theta$, Snell's law $n_1 \sin \theta_1 = n_2 \sin \theta_2$ becomes $n_1 \theta_1 = n_2 \theta_2$. Substituting the two exterior-angle relations,

$$n_1(\alpha + \beta) = n_2(\beta - \gamma) \quad \implies \quad n_1\alpha + n_2\gamma = (n_2 - n_1)\beta. \quad (8)$$

Step 3 — small-angle geometry. Each angle belongs to a skinny triangle with vertical leg h and a horizontal leg measured to the vertex (the sag of the surface is second-order and neglected):

$$\alpha \approx \frac{h}{S}, \quad \beta \approx \frac{h}{r}, \quad \gamma \approx \frac{h}{S'}. \quad (9)$$

Substituting and cancelling the common factor h ,

$$\boxed{\frac{n_1}{S} + \frac{n_2}{S'} = \frac{n_2 - n_1}{r}}. \quad (10)$$

The cancellation of h is the physical content of the result: every paraxial ray from P , no matter where it strikes the surface, crosses the axis at the *same* S' — which is precisely the condition for a sharp image. At the next order in h , S' acquires an h -dependence: spherical aberration.